Sai Krishnam Raju Mudunuri

AI/ML Engineer | Atlanta, GA
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SUMMARY

AI/ML Engineer with 3.5 years of experience building production ML and GenAI systems in industrial IoT (Honeywell) and healthcare (Infosys). Builds RAG and agent workflows on AWS, fine-tunes and serves open-source LLMs on Kubernetes, and adds the evaluation and monitoring that keep them reliable.

EXPERIENCE

Honeywell (Contract) | AI/ML Engineer **Oct 2024 – Present**
Atlanta, GA

- Built a production RAG assistant over **10,000+** internal engineering documents using Amazon Bedrock (Titan embeddings) and OpenSearch hybrid search (BM25 + vector, metadata filters, reranking, citations); cut response latency from 6 s to 2.3 s (**62%**).
- Developed LangGraph agents automating enterprise workflows across **20+** authenticated FastMCP tools and internal APIs, with scoped permissions plus schema and output validation to block invalid tool calls before execution.
- Fine-tuned Gemma with QLoRA and served the quantized model with vLLM on GPU-enabled Amazon EKS, reducing GPU memory usage by **45%** and inference costs by **40%** versus the baseline deployment.
- Built an Isolation Forest anomaly detector on industrial telemetry (temperature, pressure, vibration, motor current, flow): **0.92 ROC-AUC**, 83% precision, 91% recall; reduced false maintenance alerts by **35%** versus static thresholds.
- Implemented LLM evaluation and tracing with LangSmith, RAGAS, and DeepEval to identify hallucinations, retrieval failures, and invalid tool calls; improved response reliability through trace-level analysis, retrieval tuning, and output validation.
- Automated model releases with Terraform, Helm, Argo CD, GitHub Actions, and SageMaker Pipelines, cutting release time from 2 days to **4 hours** with canary rollouts and version-controlled, reproducible deployments.

Infosys | ML Engineer **Jan 2022 – Jul 2023**
Chennai, India

- Built XGBoost, LightGBM, and Random Forest models for readmission risk and disease progression, with hyperparameter tuning and cross-validation; best model reached **0.90 ROC-AUC** and **88% recall**, explained with SHAP.
- Built ClinicalBERT pipelines extracting diagnoses, medications, and observations from clinical notes, cutting chart review by **60%**.
- Preprocessed and engineered features from **15M+** EHR, lab, and encounter records with Python, PySpark, Pandas, and NumPy.
- Automated SageMaker deployment, versioning, and retraining with Docker, MLflow, and CI/CD, cutting deployment time by **50%**; added drift monitoring (Model Monitor, CloudWatch) that reduced model-quality incidents by 30%.

PROJECTS

Customer Churn Prediction Service (Core ML) **github.com/SaiMudunuri04/customer-churn-service**

- Built a leakage-aware scikit-learn pipeline (imputation with missing indicators, scaling, class-weighted logistic regression).
- Used chronological splits and validation-only threshold tuning; evaluated ROC-AUC, F1, precision, recall, and Brier score.
- Served churn scores through FastAPI with Pydantic feature-schema validation, health/readiness probes, and SageMaker training.
- Automated CI/CD with GitHub Actions (pytest, ruff, Helm lint), pinning tested image SHAs for Argo CD on Kubernetes.

LangGraph Customer Support Agent (GenAI) **github.com/SaiMudunuri04/langgraph-support-agent**

- Built a LangGraph state-machine copilot that routes user intent with an LLM to knowledge-base (BM25) and ticket-lookup tools.
- Added a guardrail node that validates answers (empty, length, prompt-leak checks) and escalates failures to a human handoff.
- Created a golden-set evaluation for routing and answer grounding plus 14 pytest tests, run in CI on every push.
- Served via FastAPI on Amazon Bedrock with LangSmith tracing and Secrets Manager; deployed with Docker, Helm, Argo CD.

SKILLS

GenAI & LLMs: LangGraph, LangChain, MCP/FastMCP, Bedrock, Hugging Face, vLLM, QLoRA, Gemma, LangSmith, RAGAS, DeepEval
RAG & Search: OpenSearch, FAISS, Pinecone, Redis, BM25, hybrid search, reranking
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, SHAP, ClinicalBERT
MLOps & Cloud: AWS (SageMaker, EKS, S3, CloudWatch), Docker, Kubernetes, Terraform, Helm, Argo CD, GitHub Actions, MLflow
Languages & Data: Python, SQL, PySpark, Pandas, NumPy, FastAPI, REST APIs

EDUCATION

M.S., Information System Technologies | Wilmington University, DE

B.Tech., Computer Science and Engineering | Rajalakshmi Engineering College, Chennai